NPL Yield Optimization: "Cake and Icing" Strategy

Author: Gram99
Target Stakeholders: Recovery Operations & Strategy (1st & 2nd Line of Defense)

📊 Executive Summary

This interactive dashboard is a decision-support tool designed to optimize the Net Present Value (NPV) of Non-Performing Loan (NPL) portfolios. By moving beyond binary "Sell vs. Hold" choices, this model demonstrates a **Hybrid Recovery Strategy**: isolating the high-FICO "tail" for internal/legal recovery while utilizing bulk asset sales for the lower-propensity "bedrock."

Key Business Impact:

- **Yield Enhancement:** Demonstrates how the 3-to-7 year "legal tail" can outperform 8% bulk asset sales by 2.5x on an NPV basis.
- **Operational Bandwidth:** Explicitly incorporates "Cost-to-Collect" and team capacity constraints to ensure the 1st Line (Operations) is not over-leveraged.
- **Strategic Alignment:** Directly supports Citi's 2026 Transformation goals of modernizing infrastructure and enhancing data-driven performance.

🚀 Interactive Features & Executive UX

The application is built using **Python and Streamlit**, featuring a precision-grade UI designed for senior leadership review.

Tab 1: Strategic Simulator (NPV Crossover)

- **Goal:** Identify the exact point where the time-value of legal recovery exceeds immediate cash-in-hand.

- **Precision Inputs:** Includes granular `number_input` fields for exact Discount Rates and Asset Sale Percentages (e.g., 7.25%), moving beyond the friction of traditional sliders.
- **NPV Modeling:** Uses dynamic curves to visualize the impact of the bank's cost of capital over a 7-year horizon.

Tab 2: Portfolio Lab (Segmentation)

- **Goal:** Visualize the distribution of the portfolio and define the "FICO Floor."
- **Audit-Ready Data:** The "Priority Litigation Targets" table utilizes strict two-decimal currency formatting with comma separators (e.g., $24,200.00) to mirror internal bank ledger standards.
- **Behavioral Modeling:** Incorporates a "Propensity to Pay" (P2P) score to filter high-yield targets within the chosen FICO segment.

🛠️ Technical Implementation

This project demonstrates proficiency in financial engineering, data visualization, and full-stack Python deployment.

- **Languages:** Python 3.x
- **Libraries:** Pandas (Data manipulation), NumPy (Financial modeling), Plotly (Dynamic graphing), Streamlit (Application framework).
- **Structure:**
  - `app.py` : Main dashboard application logic.
  - `requirements.txt` : Environment configuration for one-click deployment.
  - `data/` : Synthetic portfolio generation engine simulating real-world NPL behaviors.

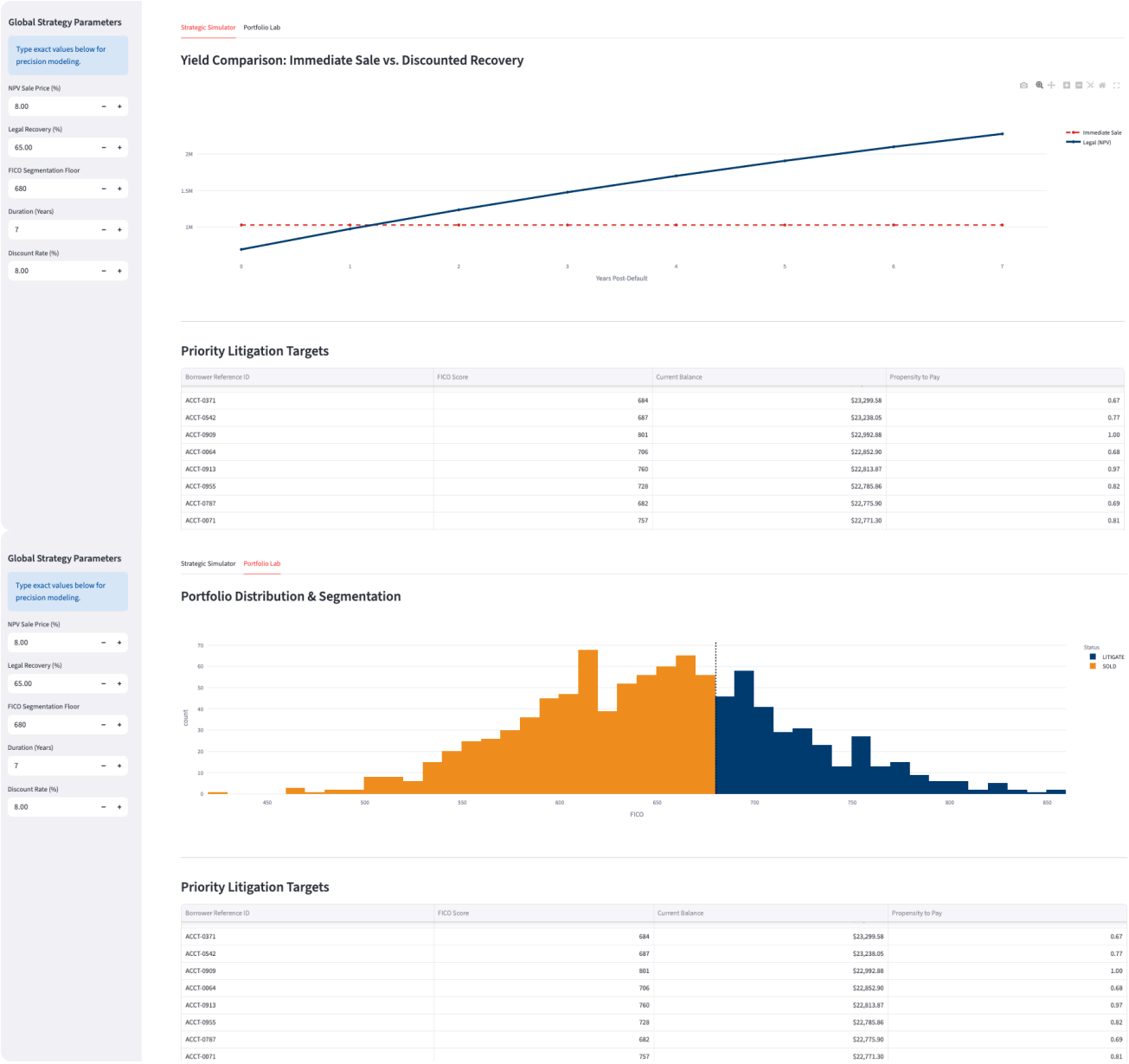
💻 Installation & Usage

To run this dashboard locally, ensure you have Python installed and follow these steps:

1. **Clone the repository:** `git clone https://github.com/your-username/recovery-legal-strategy-portfolio.git`
2. **Install dependencies:** `pip install -r requirements.txt`

3. **Launch the application:** `streamlit run app.py`

📁 Repository Strategy

This repository serves as a "narrative of authority," demonstrating that strategic ideas in the 2nd line of defense must be grounded in the operational reality of the 1st line.

Screen Shots